

DISCRETE MATHEMATICS — ULTIMATE REVISION SHEET

Key topics, formulas & quick-fire tricks for PCC-CSE 404

RELATIONS & FUNCTIONS

POSET / Equivalence

- POSET: reflexive + antisymmetric + transitive
- Equivalence rel: refl+sym+trans $\Rightarrow$ partitions set into equivalence classes

Functions

- $(f \circ g)(x) = f(g(x))$; compute g first, plug into f
- Bijective = injective + surjective $\Rightarrow$ inverse exists
- $f(x) = x^3 \Rightarrow f^{-1}(x) = \text{cube root } x$

Domain/Range tricks

- Domain: denom $\neq 0$, expr under $\sqrt{}$ ≥ 0
- Range of sin/cos combos: use $R = \sqrt{a^2 + b^2}$ bound
- Range of $x/(x^2+1)$: use AM-GM, $|f(x)| \leq 1/2$

NUMBER THEORY: DIVISIBILITY & GCD

GCD proofs (use $d = \gcd(a,b)$)

- $a = dm$, $b = dn$ with $\gcd(m,n)=1$, substitute & simplify
- $\gcd(2a+b, a+2b)$: combine to find divides $3 \cdot \gcd(a,b)$

Special forms

- $\text{Odd}^2 = 8k+1$
- Any square $= 9k$ or $3k+1$
- Any cube $= 9k$, $9k+1$, or $9k-1$
- $n = a^2 + b^2 \Rightarrow n = 4q$, $4q+1$, or $4q+2$ (never $4q+3$)

Product of consecutive integers

- Product of r consecutive naturals divisible by $r!$ (proof: equals $r! \cdot C(n+r-1, r)$, an integer)

OPHANTINE EQUATIONS & MOD ARITHMET

$ax+by=c$

- Solvable iff $\gcd(a,b) \mid c$
- Use Extended Euclidean Algo $\rightarrow$ find x_0, y_0
- General: $x = x_0 + (b/d)t$, $y = y_0 - (a/d)t$
- Linear Congruence $ax \equiv b \pmod{m}$
- Let $d = \gcd(a,m)$. Solvable iff $d \mid b$
- If solvable $\rightarrow$ exactly d solutions mod m
- Divide through by d , solve reduced eqn, then list all d solutions by adding (m/d)
- Inverse mod m
- Solve $ax \equiv 1 \pmod{m}$ via Extended Euclid
- Power/Unit digit tricks
- Find cycle of last digits (period 4 usually)
- Use Fermat's Little Thm: $a^{p-1} \equiv 1 \pmod{p}$, p prime

COMBINATORICS — PERMS & ARRANGEMENT

Repeated letters

- Total arrangements $= n!/(p_1!p_2!\dots)$
- With E first & last: fix positions, arrange rest

No-two-adjacent

- Arrange n items first ($n!$), place others in $(n+1)$ gaps using permutations

Spacing constraint ($\geq k$ between specials)

- Arrange majority group first, create gaps, insert special items respecting min-gap rule

Dictionary Rank

- For each letter (left to right): count letters smaller than it among remaining (unused) letters, multiply by (remaining positions)!/(repeats!)
- Sum all + 1 = rank

Circular arrangements

- $(n-1)!$; if reflection counts same: $(n-1)!/2$

COMBINATORICS — SELECTIONS & COUNTIN

Stars & Bars (with repetition)

- $C(n+r-1, r-1)$: r types, choose n items

Bounded solutions (Inclusion-Exclusion)

- $x_1 + x_2 + x_3 = n$, $x_i \leq k_i$:
Total $C(n+2, 2) - \sum[\text{exceed one bound}] + \dots$

Divisors of $N = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots$

- Number of divisors $= (a_1+1)(a_2+1)(a_3+1)\dots$
- Sum of divisors $= \prod[(p_i^{a_i+1}-1)/(p_i-1)]$
- $2160 = 2^4 \cdot 3^3 \cdot 5 \rightarrow 40$ divisors

Distribution into groups

- Unequal distinguishable groups: multinomial $n!/(n_1!n_2!\dots n_k!)$
- Identical-size identical groups: divide by $k!$

Inclusion-Exclusion for 'divisible by'

- $|A \cup B \cup C| = \sum A - \sum A \cap B + |A \cap B \cap C|$, use floor (N/d)

PIGEONHOLE PRINCIPLE

Basic / Generalized

- n pigeons, k holes $\Rightarrow$ some hole has $\geq \text{ceil}(n/k)$ pigeons
- Rooms needed $= \text{ceil}(\text{classes/time-slots})$

Picking numbers from $\{1..9\}$

- Sum even: pick from one parity class + 1
- Difference $= 5$: pairs $(1,6)(2,7)(3,8)(4,9)$ - 5 pairs, picking $n=6$ forces a match

Geometric Pigeonhole

- Divide region (triangle/square) into k congruent sub-regions of known max diagonal
- If points $> k$, two points fall in same sub-region $\Rightarrow$ distance $<$ diagonal
- Square side 2: divide into 4 unit squares, diagonal $= \sqrt{2} \Rightarrow 5$ points guarantee $< \sqrt{2}$
- Equilateral triangle: divide into 4 similar triangles via midpoints

PROPOSITIONAL LOGIC

Core equivalences

- $p \rightarrow q \equiv \neg p \vee q$
- $\neg(p \rightarrow q) \equiv p \wedge \neg q$
- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (p \vee q) \rightarrow (p \wedge q)$

De Morgan's & Distribution

- $\neg(p \wedge q) = \neg p \vee \neg q$ $\neg(p \vee q) = \neg p \wedge \neg q$
- $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ [distributive]

Absorption / Idempotent

- $p \wedge (p \vee q) = p$ $p \vee (p \wedge q) = p$

Normal Forms

- CNF = AND of OR-clauses
- DNF = OR of AND-clauses
- Method: remove $\rightarrow, \leftrightarrow$ using equivalences above, apply De Morgan to push $\neg$ inward, distribute

Tautology/Contradiction

- Tautology: truth table all T (e.g. $p \rightarrow (q \rightarrow p)$)
- Contradiction: all F (e.g. $(p \wedge q) \wedge \neg(p \vee q)$)

GROUPS, SUBGROUPS & CYCLIC GROUPS

Group Axioms ($G, *$)

- Closure, Associative, Identity e , Inverses exist
- Abelian = also commutative

Subgroup Test

- $H \neq \emptyset$ subset of G is subgroup iff $\forall a, b \in H: a * b^{-1} \in H$
- Intersection of subgroups = subgroup (always true)
- Union of subgroups: NOT always a subgroup

Order of elements

- $o(a) = n$ means $a^n = e$ (smallest such n)
- $a^m = e \iff n \mid m$
- $o(y^k) = n / \gcd(n, k)$ [if $o(y) = n$]

Cyclic groups

- $G = \langle a \rangle$ for some $a \in G$; ALL cyclic groups Abelian
- Generators of Z_n = elements coprime to n
- Z_6 gens: $\{1, 5\}$ | Z_9 gens: $\{1, 2, 4, 5, 7, 8\}$

Lagrange's Theorem

ISOMORPHISMS, RINGS & GRAPH THEOR

Homomorphism $f: G \rightarrow G'$

- $f(a * b) = f(a) *' f(b)$; $\text{Ker } f = \{x: f(x) = e'\}$
- Monomorphism = injective hom
- $f(x) = 4x$ on $(Z, +) \rightarrow \text{hom}$, $\text{Ker} = \{0\}$
- $f(x) = |x| \rightarrow$ check $f(a+b) = f(a) + f(b)$: fails, NOT hom

Rings & Integral Domains

- Integral domain = comm. ring + unity + no zero divisors
- Finite integral domain $\Rightarrow$ Field

Trees & Euler's Formula

- Tree: connected + acyclic; n vertices $\rightarrow n-1$ edges
- Planar connected: $V - E + F = 2$ (F = all faces incl outer)
- Binary tree: pendant vertices $= (n+1)/2$

Spanning Trees

- Prim's: grow from a vertex, add cheapest connecting edge
- Kruskal's: sort edges, add if no cycle (until $n-1$ edges)

Coloring & Matching

- Chromatic poly of tree: $k(k-1)^{n-1}$